

Problem Set 2

Total 15 points

Due Monday, 23 February 2025 before 11:59pm

Instructions: This is a GROUP assignment.

I think it makes sense to submit an excel spreadsheet if you work with excel/VBA, a word document or notebook with embedded figures if you use R/Python.

(a) Only submit one assignment per group but make it very clear who the group members are.

(b) The file name should be: PSn.FAMILYNAME1Initial1_FAMILYNAME2Initial2....xls, for example my solution to Problem set 1 would have the file name PS1.WIDDICKSM.xls.

(c) Please make sure that your solutions are well-organized and clear, with appropriate text, explanations, and formatting. If I cannot figure out what you did, then what you did is wrong.

Q1 Link between binomial model and Black-Scholes PDE [4 points]

Consider a simplified binomial model with time step of dt where $u = 1 + \sigma\sqrt{dt}$, $d = 1 - \sigma\sqrt{dt}$, (note that this is the CRR model for very small time steps dt). The option value today is $V = V(S, t)$, option value in the up-state is $V^+ = V(u \times S, t + dt)$, and option value in the down-state is $V^- = V(d \times S, t + dt)$ show that in the limit of $dt \rightarrow 0$ we recover the Black-Scholes partial differential equation (which we will see again!):

$$0 = \frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV$$

Hints. This question is asking you to carry out a small bit of analysis. Expand V^+ and V^- in Taylor series (your Taylor series needs to be second order in S and first order in t), for example.

$$V^+ = V + (uS - S) \frac{\partial V}{\partial S} + (t + dt - t) \frac{\partial V}{\partial t} + (uS - S)^2 \frac{1}{2} \frac{\partial^2 V}{\partial S^2}$$

Then substitute these expansions into the standard binomial pricing formula,

$$V(S, t) = e^{-r dt} (qV^+ + (1 - q)V^-)$$

substitute in the definition of q and **expand all of the exponential terms using $e^x = 1 + x$** , i.e ignoring smaller terms in the expansion. With any luck, you should obtain the partial differential equation above. Taylor Series will be very useful in a couple of weeks so it's useful to get some practice in!!

Q2 Practice with the Binomial (7 points)

Adapt the trees from ImpBinomial.xls to value a slightly simplified version of this product for United Parcel Services (UPS). Alternatively, adapt the python code to perform the same valuation

[Term Sheet Here](#)

I have tried to match the r , d , and σ as closely as possible (saving you time this time here) you are welcome to try to obtain this data yourself (especially the dividend history)

Face value: \$1000

Payoff at maturity: if the stock price is greater than or equal to \$91.12 then you receive the face value plus the final coupon payment. If the stock price is below \$91.12 you receive a cash amount equivalent to $1000/91.12 = 10.9745$ stocks plus the final coupon payment.

Autocall feature: If the stock price is greater than or equal to the initial price (\$107.20) on any of the Observation Dates (Apr 27, 2026, ...) then the notes are immediately called for the face value + coupon on the corresponding Coupon Payment Date (Apr 29, 2026. ...)

Coupons: there is a monthly coupon of 8.5% (this is an annual figure) of the face value, payable on the Coupon Payment Dates (Mar 3, 2026, Mar 31, 2026, ...)

To set up your tree choose:

$N = 100$ (i.e twice the size of the ones in class) or (preferably) $N =$ number of days from pricing to valuation.

$S_0 = 107.20$

$T =$ Calculate this ☺

$r = 3.443\%$ (assume the two ' r ' values are the same here)

$\sigma = 29.215\%$ (ATM) or 32.318% (85% moneyness)

Proportional Dividends: 6.415% (annual figure) quarterly on Feb 17, 2026; May 18, 2026; Aug 17, 2026; and Nov 16, 2026. You can approximate these, but make sure they are roughly correct relative to the autocall observation dates.

Use the same u , d formulas as given in the spreadsheet and python code in class (obviously adjusting for σ).

Q3 First real coding exercise – analysis of European option prices [4 points]

Consider a European put option where $S_0 = 100$, $K = 95$, $r = 0.03$, $\delta = 0$, $\sigma = 0.15$ and $T = 1$.

- a. Calculate the value of the European put options using the Black-Scholes formula.

We will now compare the performance of the following binomial lattice methods.

- Cox, Ross and Rubinstein, 1979 ($u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$)
- Rendleman and Bartter, 1979 ($u = \exp((r - \delta - 0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t})$, $d = \exp((r - \delta - 0.5\sigma^2)\Delta t - \sigma\sqrt{\Delta t})$)

For each model perform the following:

- b. Calculate the value of the European put option for time steps ranging from $N = 50$ to $N = 1000$.
- c. Plot a graph of N (x-axis) against error (y-axis) when compared to the Black-Scholes price.